Intriguing Properties of Large Language Models

In-Context Learning, Mesa-Optimizers, and the Reversal Curse

Executive Education in AI & Business

2026

Roadmap

Mechanistic Interpretability

↓

In-Context Learning & Mesa-Optimizers

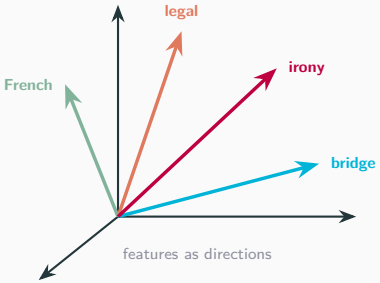
↓

OPRO: LLMs as Optimizers

The Reversal Curse

Recap: We Can Now See Inside

- Individual neurons are polysemantic
- Concepts = **directions** in activation space (superposition)
- Sparse autoencoders extract interpretable features (34M from Claude 3 Sonnet)
- Circuit tracing reveals multi-hop reasoning inside the model



Elhage et al., "Toy Models of Superposition," 2022. Templeton et al., "Scaling Monosemanticity," Anthropic, May 2024.

Part I

In-Context Learning

The model that learned to learn

The Puzzle of In-Context Learning

`J. Smith, VP Sales` → **Senior**

`A. Lee, Intern` → **Junior**

`R. Patel, CTO` → **Senior**

`M. Chen, Analyst` → **?**

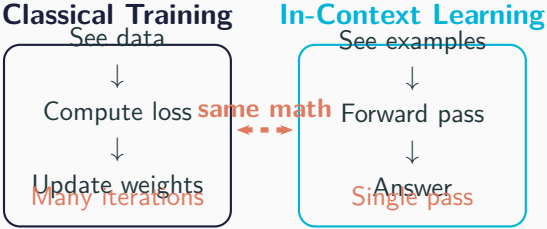
Junior

The model has never seen this classification scheme before.

No weight updates. No training.

It **inferred the rule** from three examples and **applied it**, in a single forward pass.

The Forward Pass *Is* Gradient Descent



Two independent research groups proved in 2023:

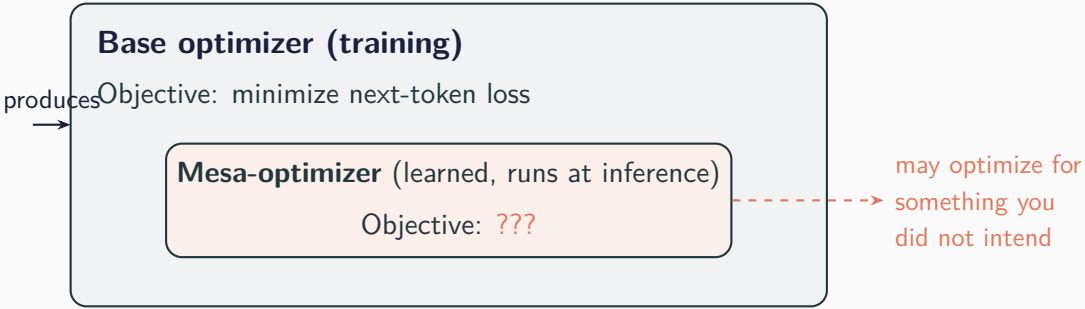
The attention mechanism **implements gradient descent** on the in-context examples.

The model does not just recall patterns. It *learns*, on the fly, without changing a single weight.

von Oswald et al., ICML 2023.

Akyurek et al., ICLR 2023.

Mesa-Optimizers: An Optimizer Inside the Optimizer



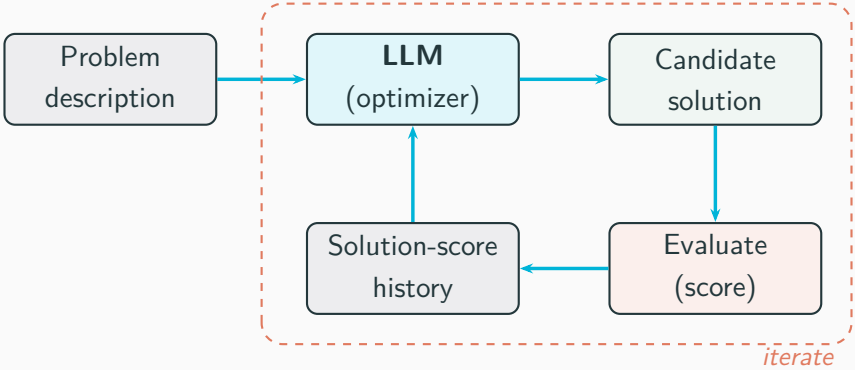
Hubinger et al., "Risks from Learned Optimization," MIRI, 2019. NeurIPS 2024: empirical confirmation.

Part II

OPRO: LLMs as Optimizers

When you hand the learning algorithm the steering wheel

OPRO: Optimization by PROmpting



Yang et al., "Large Language Models as Optimizers," ICLR 2024.

OPRO's Killer Application: Prompt Optimization

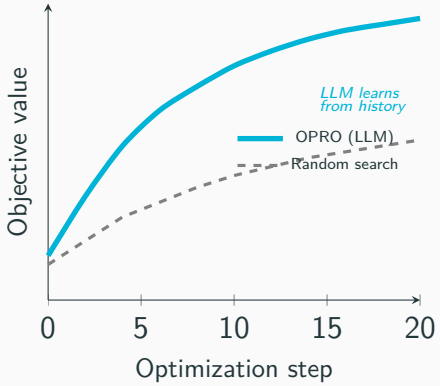
Human-designed: "Let's think step by step."
GSM8K: 71.8%

OPRO-discovered: "Take a deep breath and work on this problem step-by-step." GSM8K: **80.2%**

OPRO-discovered (PaLM 2): "Break this down."
Beats hand-crafted prompts

The optimizer discovers prompts no human would have written, and they *work better*.

OPRO Beyond Prompts: Mathematical Optimization

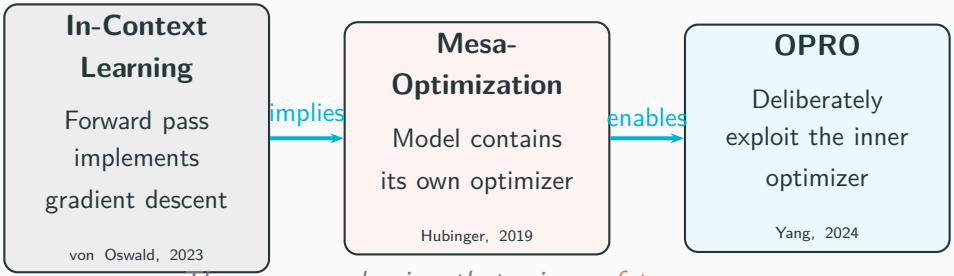


OPRO applied to combinatorial and continuous optimization:

- **Linear regression**: finds near-optimal solutions
- **Traveling salesman**: competitive with heuristics
- No domain-specific algorithm needed, just natural language

The LLM reads previous attempts and improves.

The Deep Connection



*The same mechanism that raises safety concerns also enables **powerful new capabilities**.*

A Question Worth Sitting With

If a model can learn from examples at inference time, without updating any of its parameters,

is it still "just" a model?

Or has it become a **learning algorithm** that happens to be stored in a neural network?

Part III

The Reversal Curse

A cold shower

A Question a Four-Year-Old Can Answer

Tom Cruise's mother is **Mary Lee Pfeiffer**.

Who is Mary Lee Pfeiffer's son?

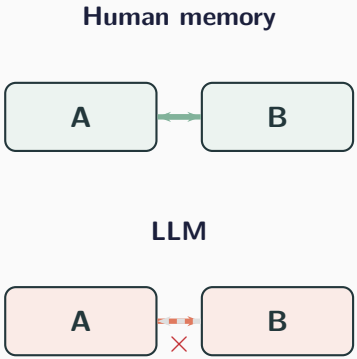
You: "Tom Cruise"

GPT-4: *"I don't know"*

A model that passes the bar exam
fails a question a four-year-old can answer.

Berglund et al., "The Reversal Curse," ICLR 2024.

Why: Gradient Updates Are Asymmetric



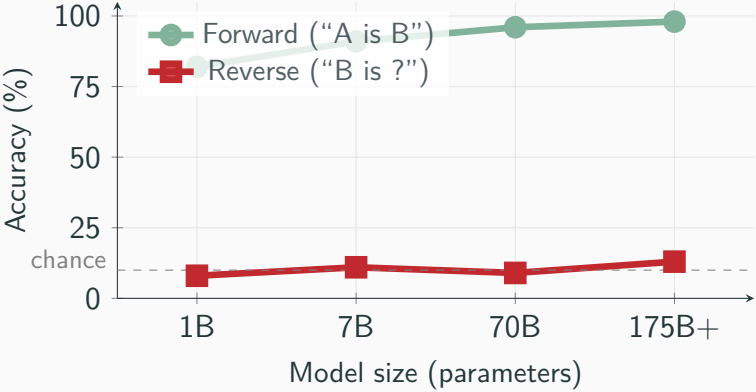
Training on "A is B":

1. Process tokens left to right
2. Gradient updates A's representation to predict B
3. B's representation is not updated to encode A

Knowledge stored as a **one-way street**.

"Is the Reversal Curse a Binding Problem?" arXiv:2504.01928, 2025.

More Parameters, Same Problem



Forward accuracy: near-perfect. Reverse accuracy: flat at chance.

Data augmentation helps partially. No real fix exists.

What This Means

1. Benchmark scores are misleading. High percentiles do not imply relational understanding.

2. LLM "knowledge" is **directional**, not relational. More like a hash table than a database.

3. For any query direction *not* in the training data, the model has systematic blind spots.

4. Despite passing exams, the model's cognition is **fundamentally alien** to ours.

The Paradox Inside the Black Box

It learns at inference time

Internal gradient descent, OPRO

Astonishing

It can be used as an optimizer

Mesa-optimization made productive; also a safety risk

Powerful & dangerous

Its knowledge is one-way

Reversal curse at all scales; no known fix

Broken

"Sophisticated enough to optimize itself. Too alien to reverse a simple fact."

Discussion

1. OPRO shows that LLMs can optimize their own instructions better than humans can. What does this mean for the **role of prompt engineers**?

2. A model that learns from every prompt at inference time might behave differently depending on context. How should organizations **test and govern** a system whose behavior is context-dependent by design?

3. The Reversal Curse persists at all model sizes. If you were deploying an LLM for a **knowledge-intensive** task (medical diagnosis, legal research), what safeguards would you require?

Key References

In-Context Learning

von Oswald et al., "Transformers Learn In-Context by Gradient Descent," ICML 2023.

Akyurek et al., "What Learning Algorithm Is In-Context Learning?" ICLR 2023.

Olsson et al., "In-Context Learning and Induction Heads," Anthropic, 2022.

Mesa-Optimizers

Hubinger et al., "Risks from Learned Optimization," MIRI, 2019.

"On Mesa-Optimization in Autoregressively Trained Transformers," NeurIPS 2024.

OPRO

Yang et al., "Large Language Models as Optimizers," ICLR 2024.

The Reversal Curse

Berglund et al., "The Reversal Curse," ICLR 2024.

"Is the Reversal Curse a Binding Problem?" arXiv:2504.01928, 2025.